



ELECTROHACK 2.0 2026

PROJECT TITLE: MedScan

GROUP: Hacker Berjaya

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1. Problem Statement

- Patients, particularly the elderly or those with visual impairments, often struggle to read and interpret the small print on medication packaging. Most manufacturers use font sizes small as 4-point or 6-point to fit extensive regulatory requirements on a tiny bottle or box.
- Current medication packaging relies heavily on dense text and complex medical jargon, making it difficult for the average consumer to quickly understand critical information. This way of information is structured in and phrased often presents a significant cognitive barrier, particularly for individuals with low health literacy.
- Caregivers and healthcare workers frequently need to identify loose or unboxed medications quickly. In addition, hundreds of different medications share nearly identical physical characteristics. It becomes incredibly easy to mistake each other.

2. How the Project Work

Phase 1

- Camera initialization occurs, and the web app requests access to the device's camera.
- The user is provided with an option to upload a photo to bypass the live camera feed.
- **SOS Feature:** The user interface renders a highly visible, persistent "SOS Emergency" button designed for high accessibility (large size, high contrast).

Phase 2

- The web application transfers the current frame of the video onto a hidden HTML canvas.
- Data Extraction takes place, where the image is converted and extracted in Base64 format.
- **SOS Feature:** If the user presses the SOS button, the system immediately registers the click event, pausing the camera analysis to prioritize generating an emergency payload (which can include a timestamp or predefined patient ID).

Phase 3

- The Base64 image data is bundled with a highly specific prompt, instructing the AI to act as a medical assistant.
- This payload is sent via a POST request directly to the Gemini API endpoint (implemented in pure HTML/JavaScript, acknowledging the absence of a .env file).
- **SOS Feature:** The emergency event triggers a separate, immediate client-side POST request. Since it is pure HTML, this is routed through a web

service like a Telegram Bot API or EmailUS to push a direct notification to the healthcare assistant's device.

Phase 4

- Once the Gemini API completes its analysis, the text response travels back to the application to be displayed on the screen.
- The audio system is triggered, reading aloud the name and the usage of the medicine for visually impaired users.
- **SOS Feature:** The UI immediately displays a clear confirmation message (e.g., "Help is on the way"), and a distinct audio prompt is triggered to reassure the user that the healthcare assistant has been successfully notified.

3. Features/functions:

1. Scan medicine using camera

- scanned images will be processed to identify the name of the drugs
- allows users to use their device's webcam or smartphone camera to capture a live image of a pill

2. Provide information on the medicine

- Displays structured information extracted during the AI scan, breaking down complex medical jargon into simple terms.

3. Text-to-Speech

- Takes the text results from the screen and reads them out loud to the user
- Creating a fully accessible experience for users who cannot see the screen

4. Emergency SOS

- button to directly go to whatsapp for emergency contact
- faster for emergency

5. Enlarge Text for elderly

- for elderly with low vision

4. Technology Used:

Web APIs :

- MediaDevices API (navigator.mediaDevices): Used to detect available cameras
- Canvas API: Used to draw the live video frame onto a 2D plane
- File API (FileReader): Allows the browser to read the contents of an uploaded image file directly on the user's device.
- Fetch API: The modern standard for making HTTP network requests, used here to communicate with the AI model.

The Backend :

- Google Gemini API: Instead of a traditional database, the application acts as an API client for Google's Generative AI.
- **Model:** Specifically, it targets the gemini-2.5-flash model, which is a multimodal LLM capable of processing both images and text simultaneously.

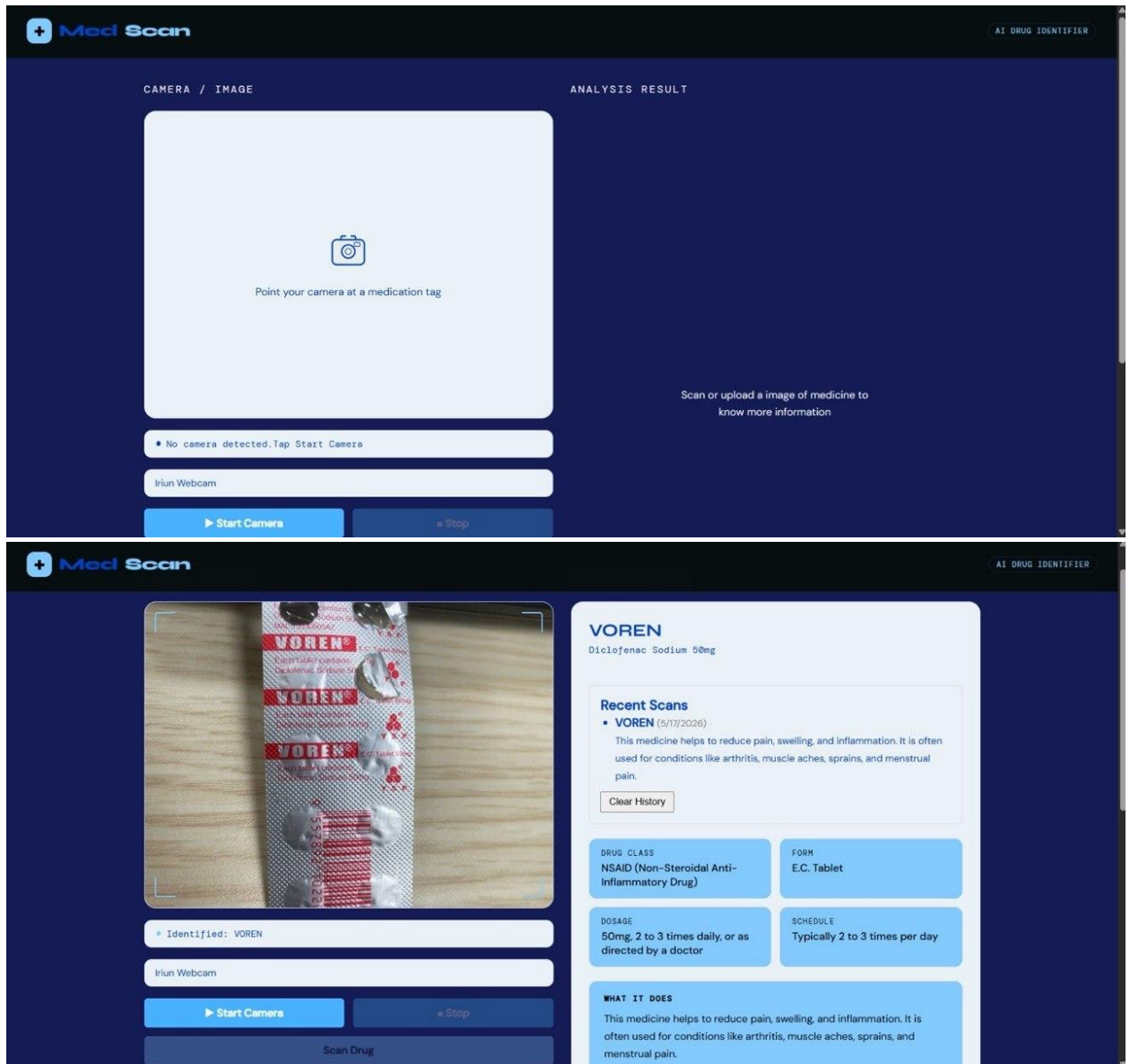
Frontend Core :

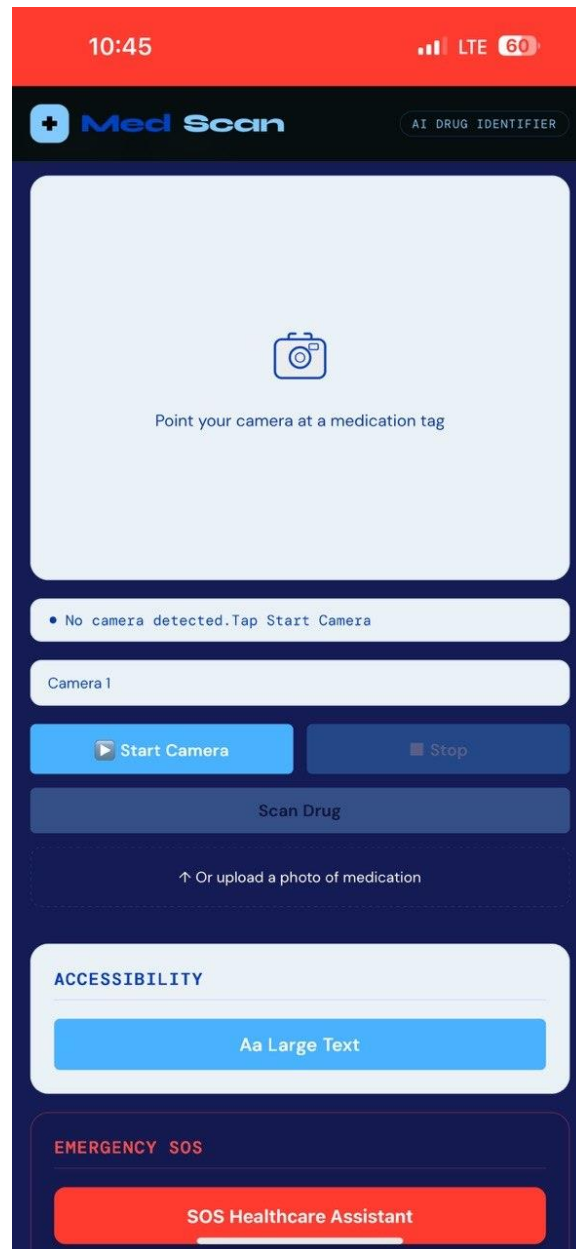
- HTML5: Used for the semantic structure of the application.
- CSS3: Handles all styling, layout, and visual feedback without relying on external libraries
- JavaScript: Controls all the logic, user interactions, and DOM manipulation.

5. Improvement:

1. Collab with KKM to ensure the trueness of the medical suppliance.
2. The budget for our project is around 500k to be published.
3. Repeated Text-to-Speech as sometimes we cannot catch up when first hear it.

6. Attachment





Github Link:

<https://github.com/syahmiii05/Electrohack>